

```
import os
print(os.listdir('/content'))

['.config', 'spam.csv', 'sample_data']
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, classification_report, confusion
```

```
df = pd.read_csv('/content/spam.csv', encoding='latin-1')

df.head()
```

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4	
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN	
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN	
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN	
U dun sav so early hor... U c						

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
print("Dataset Shape:", df.shape)
print("Columns:", df.columns)
```

```
Dataset Shape: (5572, 5)
Columns: Index(['v1', 'v2', 'Unnamed: 2', 'Unnamed: 3', 'Unnamed: 4'], dtype=
```

```
df = df[['v1', 'v2']]
```

```
df = df.rename(columns={
    'v1': 'label',
    'v2': 'message'
})

df.head()
```

	label	message
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
print(df.isnull().sum())
```

```
label      0
message    0
dtype: int64
```

```
df = df.drop_duplicates()

print("New Shape:", df.shape)
```

```
New Shape: (5169, 2)
```

```
print(df['label'].value_counts())
```

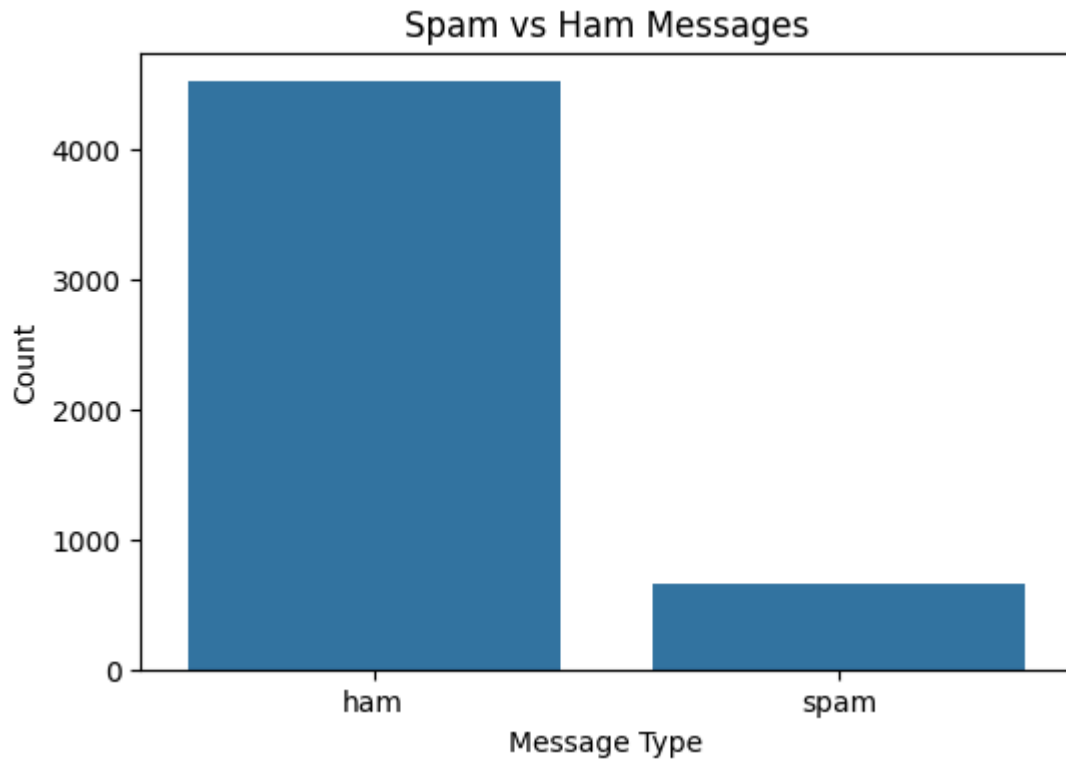
```
label
ham      4516
spam      653
Name: count, dtype: int64
```

```
plt.figure(figsize=(6, 4))

sns.countplot(data=df, x='label')

plt.title('Spam vs Ham Messages')
plt.xlabel('Message Type')
plt.ylabel('Count')

plt.show()
```



```
df['label_num'] = df['label'].map({
    'ham': 0,
    'spam': 1
})

df.head()
```

	label	message	label_num	
0	ham	Go until jurong point, crazy.. Available only ...	0	
1	ham	Ok lar... Joking wif u oni...	0	
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	1	
3	ham	U dun say so early hor... U c already then say...	0	
4	ham	Nah I don't think he goes to usf, he lives aro...	0	

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
print(df[['label', 'label_num']].head())
```

```
  label  label_num
0   ham          0
1   ham          0
2  spam          1
3   ham          0
4   ham          0
```

```
X = df['message']  
y = df['label_num']
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X,  
    y,  
    test_size=0.20,  
    random_state=42,  
    stratify=y  
)  
  
print("Training Data:", len(X_train))  
print("Testing Data:", len(X_test))
```

```
Training Data: 4135  
Testing Data: 1034
```

```
vectorizer = TfidfVectorizer(  
    lowercase=True,  
    stop_words='english'  
)  
  
X_train_tfidf = vectorizer.fit_transform(X_train)  
  
X_test_tfidf = vectorizer.transform(X_test)  
  
print("Training Matrix Shape:", X_train_tfidf.shape)  
print("Testing Matrix Shape:", X_test_tfidf.shape)
```

```
Training Matrix Shape: (4135, 7329)  
Testing Matrix Shape: (1034, 7329)
```

```
model = MultinomialNB()  
  
model.fit(X_train_tfidf, y_train)  
  
print("Model Training Completed Successfully!")
```

```
Model Training Completed Successfully!
```

```
y_pred = model.predict(X_test_tfidf)  
  
print(y_pred[:20])
```

```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]
```

```
accuracy = accuracy_score(y_test, y_pred)  
  
print("Model Accuracy:", accuracy)  
print("Accuracy Percentage:", accuracy * 100, "%")
```

Model Accuracy: 0.9661508704061895
 Accuracy Percentage: 96.61508704061895 %

```
print(classification_report(
    y_test,
    y_pred,
    target_names=['Ham', 'Spam']
))
```

	precision	recall	f1-score	support
Ham	0.96	1.00	0.98	903
Spam	0.99	0.74	0.85	131
accuracy			0.97	1034
macro avg	0.98	0.87	0.91	1034
weighted avg	0.97	0.97	0.96	1034

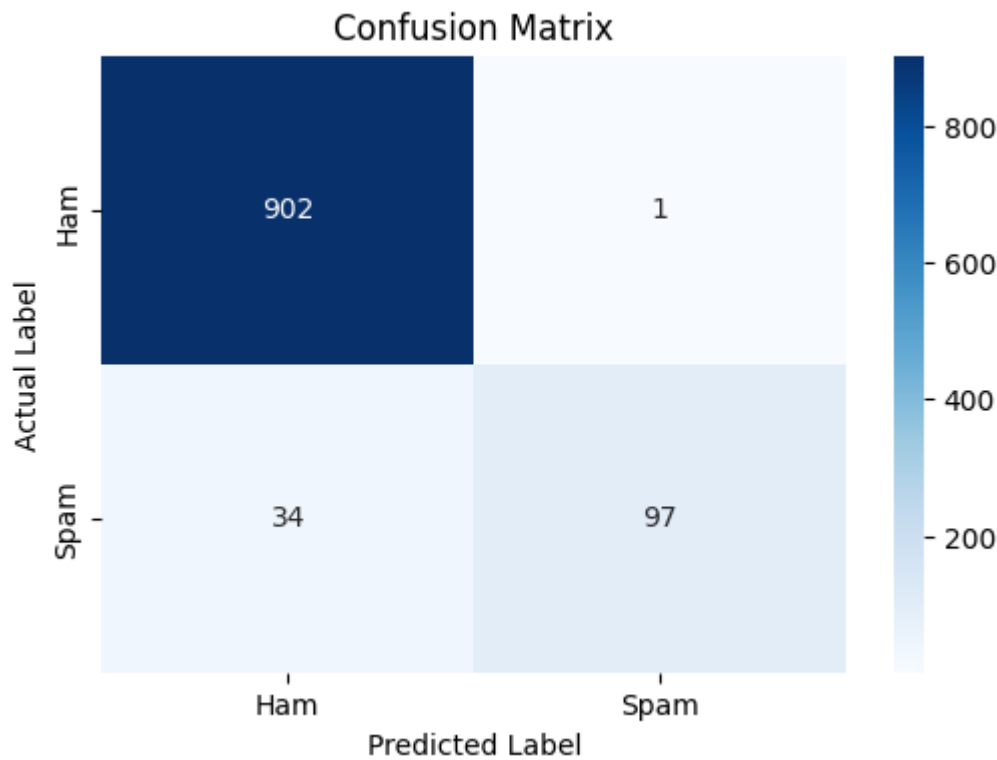
```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))

sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['Ham', 'Spam'],
    yticklabels=['Ham', 'Spam']
)

plt.title('Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')

plt.show()
```



```
def predict_spam(message):  
    message_tfidf = vectorizer.transform([message])  
    prediction = model.predict(message_tfidf)[0]  
  
    if prediction == 1:  
        return "Spam Message"  
    else:  
        return "Ham Message"
```

```
print(predict_spam("Congratulations! You won a free lottery prize."))  
print(predict_spam("Hey, are you coming to college tomorrow?"))  
print(predict_spam("You have won a cash reward. Click now to claim."))  
print(predict_spam("Please send me today's assignment."))
```

```
Spam Message  
Ham Message  
Spam Message  
Ham Message
```

```
message = input("Enter your message: ")  
  
result = predict_spam(message)  
  
print("Prediction:", result)
```

```
Enter your message: You won a free iPhone. Click this link now.  
Prediction: Ham Message
```

✓ Conclusion

In this project, a Machine Learning model was developed to classify messages as Spam or Ham. The SMS Spam Collection dataset was used for training and testing. TF-IDF Vectorization was applied to convert text into numerical features, and Multinomial Naive Bayes was used for classification.

The model was evaluated using accuracy, classification report, and confusion matrix. The project demonstrates how Natural Language Processing and Machine Learning can be used for spam message detection.

```
print("Spam Email Detection Project")
```

```
Spam Email Detection Project
```

✓ Conclusion

Our model can classify messages as Spam or Ham.

```
message = input("Enter your message: ")
```

```
result = predict_spam(message)
```

```
print("Prediction:", result)
```

```
Enter your message: Congratulations! You won a free prize. Click now!  
Prediction: Spam Message
```

```
message = input("Enter your message: ")
```

```
result = predict_spam(message)
```

```
print("Prediction:", result)
```

```
Enter your message: Hello, how are you today?  
Prediction: Ham Message
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
Accuracy: 0.9661508704061895
```